Long-Duration Verification of ManifoldsX: A Ten-Case Benchmark Study

Yosi BZ
ManifoldsX

Benchmark family	Numerical challenge	Decisive observable
Euler / Lagrange	Long-time orientation	Phase + attitude
Four-bar	Near-singular closed loop	Drive torque + closure
Kovalevskaya	Fully coupled top dynamics	Extra invariant + attitude
Duffing / pendulums	Nonlinearity and separatrix	Phase + geometry
Kepler	Concentrated pericenter dynamics	Orbit position + energy
Chain / beam	Many DOFs and multiple modes	Dispersion + modal response

Fig. 1. Why the ten cases are complementary: each isolates a different numerical weakness and requires an observable that can reveal it.

Abstract—A ten-case benchmark study evaluates long-duration trajectory accuracy, invariant preservation, constraint satisfaction and runtime of the ManifoldsX multibody dynamics solver. Each family includes a 1,000 s simulation. Eight families use exact analytical trajectory references; a general Kovalevskaya top uses exact invariants and a refined numerical trajectory, while a flexible beam uses the analytical modal solution of its linearized discrete model. Projected HEDOP5 has a passing recorded setting in every family. Representative successful settings deliver 3.10–83.10 times real-time throughput. HHT with $\alpha = 0.1$ also passes several cases, including the near-toggle four-bar; long-duration phase-sensitive cases require refinement or remain outside their accuracy gates at the tested settings. Failed and partial runs are retained. The results establish reproducible verification for the specified models and settings.

Index Terms—Multibody dynamics, analytical verification, HHT, HEDOP5, long-time integration, constraint projection.

I. INTRODUCTION

Long simulations can satisfy constraints and conserve energy closely while losing trajectory phase. A useful verification study must therefore compare independent trajectories and reactions, alongside invariants and execution time. This report consolidates the recorded ten-case suite in [1]–[6].

The suite spans free rigid-body tumbling, constrained heavy tops, a nearly singular closed loop, nonlinear oscillation, near-separatrix motion, central-force orbital motion, a large spring-mass system and flexible multibody dynamics. The physical solver evolves each model; reference trajectories are used only for scoring.

II. EXPERIMENTAL DESIGN

All reported long-run settings target 1,000 s. HHT uses $\alpha = 0.1$, the index-3 formulation, fixed steps and no position or velocity projection. Projected HEDOP5 uses the production explicit integration path with constraint projection. Local error adaptation and step cutbacks are disabled. Completion is scored separately from accuracy.

The Newton absolute acceleration tolerance is 10^{-9} and the position tolerance is 10^{-10}. The remaining-five runner allows 30 Newton iterations. Practical HEDOP5 projection triggers and tolerances are 10^{-11}; earlier strict-projection controls are retained separately.

This is an IEEE-style technical report of existing benchmark evidence. No new simulations were performed to prepare the report. Source records and plotting data were taken from the recorded evidence described in Section VII.

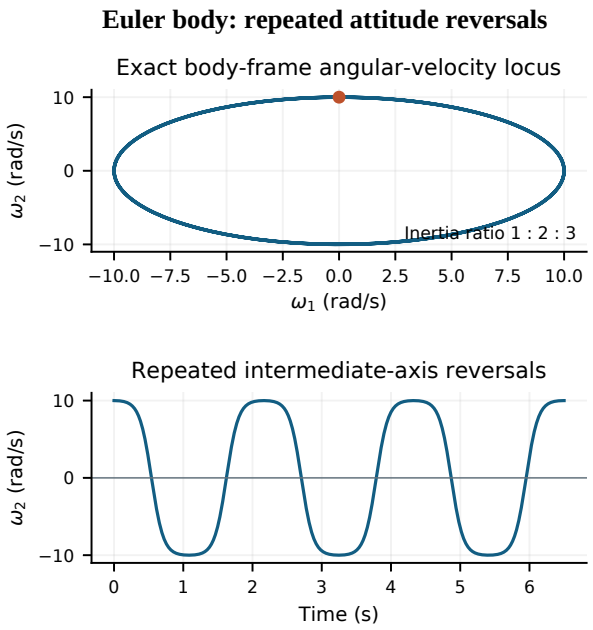


Fig. 2. Exact Euler reference: body-frame angular-velocity locus and repeated sign changes of the intermediate-axis component. No solver trajectory is used in this illustration.

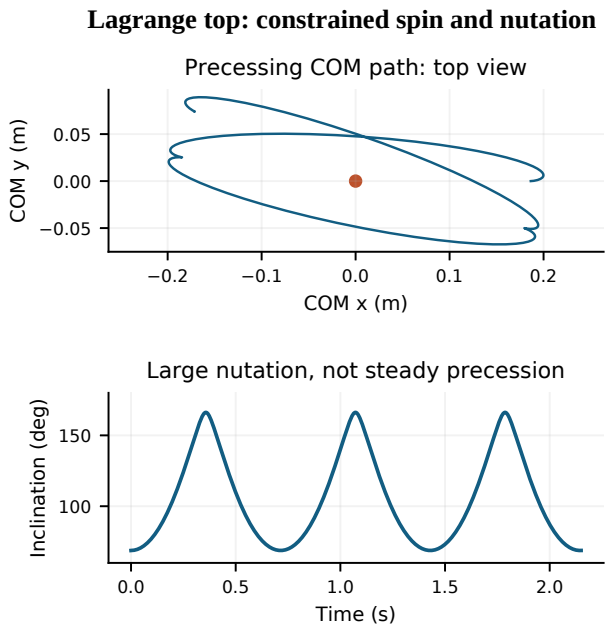


Fig. 3. Exact Lagrange-top reference: COM path in plan and inclination over approximately three nutation periods. The pivot is the origin of the plan view.

Why this case matters

A nearly intermediate-axis spin repeatedly overturns the body without external torque. Small integration errors alter when each reversal occurs; over hundreds of reversals, a visually plausible tumble can have the wrong orientation. This probes gyroscopic coupling and rotation updates without contact, forcing or joint errors obscuring the diagnosis. The same numerical ingredients arise in free-flight attitude prediction and spinning machinery.

Model and measured evidence

A torque-free body has principal COM inertia diag(1, 2, 3) kg m² and initial body angular velocity (0, 10, 1) rad/s. The exact angular-velocity solution uses Jacobi elliptic functions; attitude reconstruction uses elliptic integrals of the third kind [1], [7]. The angular-velocity period is 2.16421223445 s. A 1,000 s run includes approximately 462 periods and 924 intermediate-axis sign reversals.

Attitude is scored by the geodesic rotation difference at equal times. At 1 ms, HHT reaches 124.942° maximum error despite only 0.002784% maximum energy change. At 0.25 ms it passes with 0.626899°. HEDOP5 at 4 ms yields 0.0222071° and 69.360× RT.

What the result demonstrates

The important result is not merely that HEDOP5 finishes. It keeps maximum orientation error to 0.0222° over about 924 reversals at 4 ms. The HHT refinement shows that its long-time error is controllable by step size; the 1 ms energy result alone would have missed the trajectory failure.

Why this case matters

This top combines gravity torque, a fixed spherical pivot, spin, precession and large nutation. It exercises translational and rotational coupling and pivot reactions, while an exact attitude reference exposes phase errors hidden by tight joint closure. It is a useful building block for verifying gimballed and pivot-supported rotating assemblies. Unlike a steady-precession test, the inclination sweeps almost 100 degrees.

Model and measured evidence

The symmetric top has mass 1 kg, COM offset 0.2 m, COM inertia diag(0.02, 0.02, 0.01) kg m², initial inclination 1.2 rad and axial spin 10 rad/s. Initial nutation and precession rates are zero. A cubic first integral in cos θ gives the nutation, with elliptic integrals reconstructing precession and spin [1], [8].

The inclination spans 68.7549°–166.2315°. The 0.714715683779 s nutation period gives approximately 1,399 cycles. HHT at 1 ms has 24.4613° maximum attitude error; 0.25 ms reduces it to 0.609807°. Projected HEDOP5 at 4 ms gives 0.0200826° at 55.555× RT.

What the result demonstrates

The passing HEDOP5 result preserves full orientation over roughly 1,399 nutation cycles. HHT also passes after refinement. This separates a phase-accuracy limitation at the baseline step from an inability to represent the constrained physics.

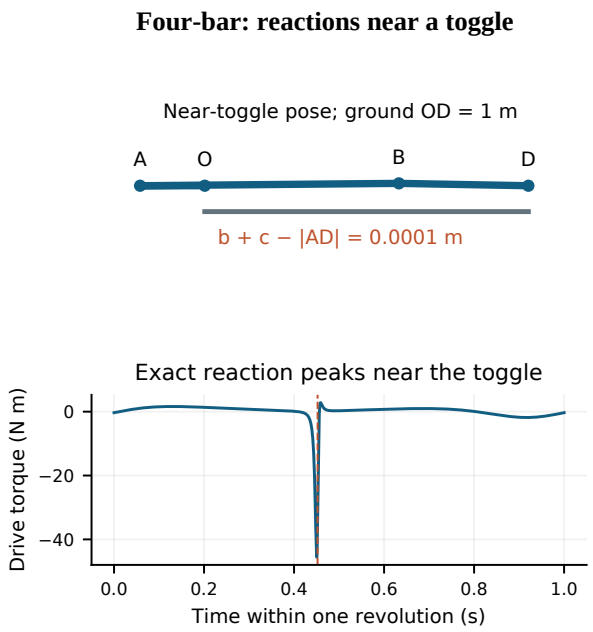


Fig. 4. Exact four-bar geometry at minimum toggle clearance and drive torque over one revolution. The near-toggle geometry is drawn to scale; the very small offset is therefore barely visible.

Why this case matters

As the coupler and rocker approach alignment, the loop-closure equations become poorly conditioned. Small kinematic errors can produce large acceleration and reaction errors even when the linkage looks correct. The drive-torque history therefore tests the constraint Jacobian, acceleration-level terms and reaction recovery together. This directly targets numerical difficulties encountered in toggle clamps, presses and near-dead-center linkages.

Model and measured evidence

Link lengths are 1, 0.1999, 0.8 and 0.4 m for ground, crank, coupler and rocker. Moving masses are 1, 2 and 1.5 kg. The initial crank angle is 0.3 rad and its prescribed speed is 2π rad/s. Gravity is disabled. The maximum separation of the coupler/rocker pivots is 1.1999 m, leaving 0.1 mm clearance from the 1.2 m toggle limit.

The exact reference uses a positive circle-intersection branch and twice-differentiated loop closure. Drive torque follows from the kinetic-energy rate. This tests reactions as well as kinematics over 1,000 revolutions. At 1 ms, HHT passes with 1.78198% peak-normalized torque error; HEDOP5 passes with 0.00357916%.

What the result demonstrates

Passing torque as well as position is the key outcome. HHT and projected HEDOP5 both meet the torque gate at 1 ms. The strict-projection and 4 ms failures show that this near-singular case also probes the practical robustness of constraint enforcement.

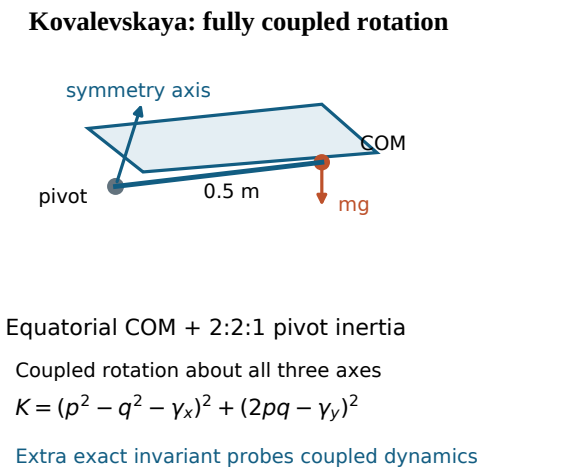


Fig. 5. Kovalevskaya model geometry (schematic) and its additional exact invariant. The COM offset lies in the equatorial plane, unlike the axial offset of the Lagrange top.

Why this case matters

Moving the COM into the equatorial plane prevents the simplifications of the symmetric heavy top with an axial COM. All three angular-velocity components interact. The extra Kovalevskaya invariant is a stringent independent diagnostic: preserving total energy alone need not preserve this fourth-degree combination. The case probes inertia-frame transformations, off-axis gravity torque and full three-axis rotational coupling.

Model and measured evidence

The unit-mass top has pivot inertia diag(2, 2, 1) kg m², gravity (0, 0, −2) m/s² and equatorial COM offset (0.5, 0, 0) m. Initial orientation is $R_y(0.7)R_z(0.4)$; body angular velocity is (0.6, 1.1, 0.8) rad/s [4].

With $\gamma = R^T e_z$ and $\omega = (p, q, r)$, exact invariants are

$E = p^2 + q^2 + r^2/2 + \gamma_x$,
$L_z = 2p\gamma_x + 2q\gamma_y + r\gamma_z$,
$K = (p^2 - q^2 - \gamma_x)^2 + (2pq - \gamma_y)^2$.

The invariant derivatives vanish symbolically. Independent Euler–Poisson DOP853 trajectories, refined in step and tolerance, differ by 1.92×10^{-9} degrees. A closed-form hyperelliptic evaluator is not implemented.

All three settings pass. HHT at 1 ms gives 0.287617° attitude error. HEDOP5 at 4 ms gives 8.72905×10^{-8} degrees and relative K drift 2.9737×10^{-13}, at 83.095× RT.

What the result demonstrates

The simultaneous small attitude error and small K drift support the coupled dynamics implementation for this trajectory. The attitude result is qualified by a refined numerical reference; the invariant identities themselves are exact.

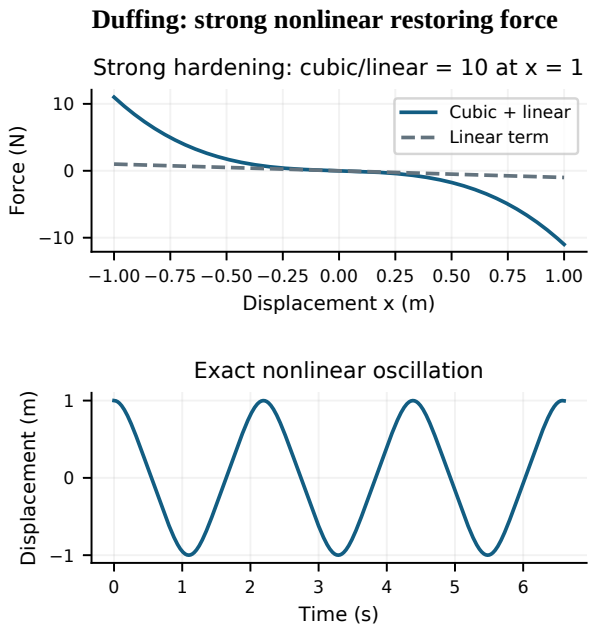


Fig. 6. Duffing restoring force compared with its linear part, and exact elliptic-function displacement. The force variation makes this a strong nonlinear oscillator.

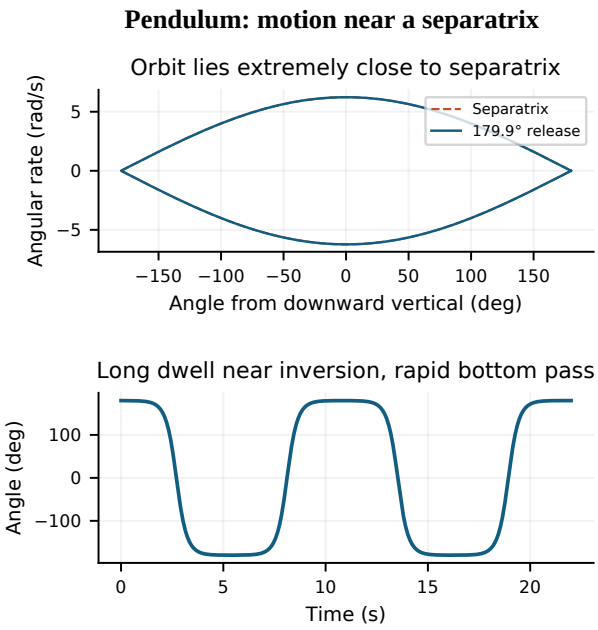


Fig. 7. Exact near-inverted pendulum phase portrait and angle history. The libration lies so close to the separatrix that the two curves almost coincide at this scale.

Why this case matters

The restoring stiffness varies strongly with displacement: its tangent is $1 + 30x^2$, rather than a constant. The oscillation frequency depends on amplitude, so numerical damping can also create frequency and phase error. This case isolates the nonlinear force law and its tangent from complicated geometry. Those ingredients are relevant to nonlinear bushings, elastic stops and hardening structural responses.

Model and measured evidence

A unit-mass slider satisfies $\ddot{x} + x + 10x^3 = 0$ with $x(0) = 1$ m and $\dot{x}(0) = 0$. A translational joint supplies five constraints. The exact solution is

$x(t) = \mathrm{cn}(\sqrt{11}\, t \mid 5/11)$,
$E = \dot{x}^2/2 + x^2/2 + 2.5x^4$.

Here the second elliptic-function argument is the parameter, not the modulus. The period is 2.19182911273 s, giving 456.24 cycles [5], [9]. HHT at 1 ms has 3.511 mm maximum displacement error and 0.23518° last-crossing phase lag. HEDOP5 at 4 ms has 0.832 µm error and 5.5655×10^{-5} degrees phase lag. All three settings pass; constraint residuals are zero.

What the result demonstrates

The measured submicrometer HEDOP5 error at 4 ms after 456 cycles supports accurate nonlinear phase evolution. HHT also passes its gates. This is force-law and integration verification; it does not establish a calibrated material law.

Why this case matters

The initial state lies extremely close to the boundary between libration and rotation. Near the inverted equilibrium, motion slows and the period becomes highly sensitive to energy. A tiny energy perturbation can therefore cause substantial crossing-time error over many swings. This is a particularly discriminating test of long-time accuracy around unstable equilibria and near snap-through or over-center configurations.

Model and measured evidence

The pendulum has mass 1 kg, COM-pivot distance 1 m, COM inertia 0.01 kg m² and pivot inertia 1.01 kg m². It is released from rest at 179.9° from downward vertical under gravity 9.81 m/s². With $\Omega = \sqrt{(9.81/1.01)}$, $k = \sin(\theta_0/2)$, the reference is

$\theta(t) = 2\,\mathrm{asin}[k\,\mathrm{sn}(K(k^2) - \Omega t \mid k^2)]$.

Its period is 10.8199908437 s. Initial energy lies only 1.49415×10^{-5} J below the separatrix. The reference is checked against 60-digit elliptic-function evaluation at late bottom crossings [6], [9]. HHT at 1 ms completes but reaches 180° bob-direction error. HEDOP5 passes at both 1 and 4 ms, with errors 0.00244311° and 0.735074°, respectively.

What the result demonstrates

The 1 ms HHT result reaches 180° direction error while its maximum energy change remains only 0.00568%. HEDOP5 passes at 4 ms and has a much larger accuracy margin at 1 ms. This case most sharply demonstrates why an energy-only acceptance test is insufficient.

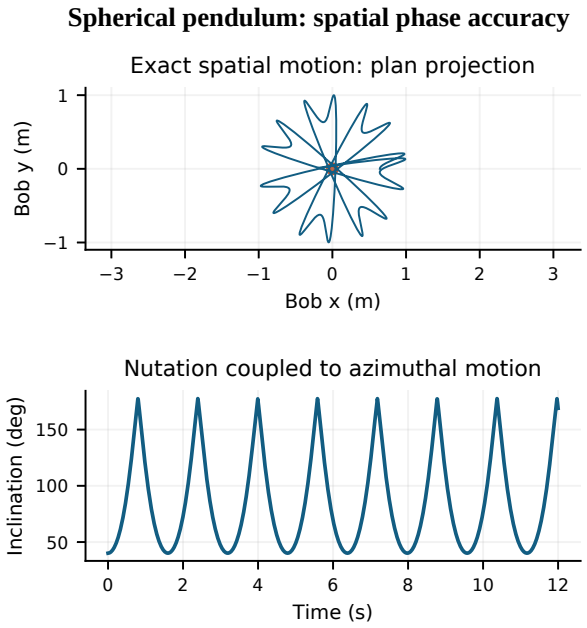


Fig. 8. Exact spherical-pendulum reference: plan projection of the bob path and inclination from upward vertical. These display spatial precession and nutation together.

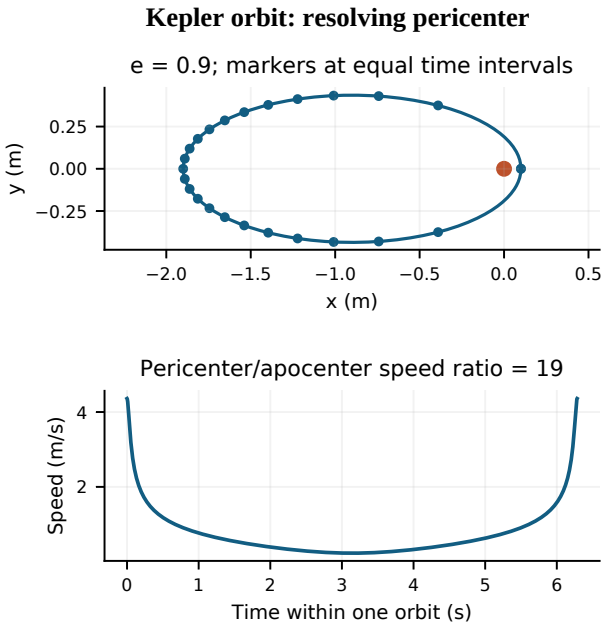


Fig. 9. Exact eccentric orbit with equal-time markers and its speed history. Sparse markers near pericenter indicate fast traversal; the central force acts from the marked focus.

Why this case matters

A planar pendulum cannot reveal errors in the coupling between azimuth and inclination. This spatial case does: its path precesses while the bob changes elevation, under a spherical-joint constraint and with finite body inertia. It tests three-dimensional geometry, angular momentum and phase simultaneously. The underlying ingredients appear in suspended payloads and spherical-joint robotic assemblies.

Model and measured evidence

A spherical pivot supports the same mass, length and transverse inertia. Initial inclination is 0.7 rad from upward vertical, azimuth rate is 0.6 rad/s, and initial nutation and axial spin are zero. A cubic first integral in $u = \cos\theta$ gives the inclination through Jacobi functions; third-kind elliptic integrals give the azimuth. The reference includes finite body inertia.

Constructing cubic roots and small root differences at 60 digits gives agreement within 3.33×10^{-12} m in five long-time reference checks. HHT at 1 ms misses the 1° direction gate with 3.30747° error; refinement to 0.5 ms passes at 0.23584°. HEDOP5 at 4 ms gives 4.13072×10^{-5} degrees at 58.669× RT.

What the result demonstrates

The 0.5 ms HHT refinement and 4 ms HEDOP5 result both pass. A spatial trajectory reference is essential: neither joint residuals nor energy alone establish that the bob occupies the correct point on the sphere at a given time.

Why this case matters

A highly eccentric orbit concentrates rapid motion and strong force gradients into a small portion of every cycle. Here the speed changes by a factor of 19 between apocenter and pericenter. Fixed steps must resolve this short demanding interval without accumulating orbital phase error across roughly 159 revolutions. The case also tests the inverse-square force, potential and analytic force derivatives.

Model and measured evidence

Mass, gravitational parameter and semimajor axis are unity; eccentricity is 0.9. Initial position is (0.1, 0, 0) m and velocity is $(0, \sqrt{19}, 0)$ m/s. The production force is $-r/|r|^3$. Solving $E - e\sin E = t$ modulo 2π gives the eccentric anomaly, then exact position and velocity. The period is 2π s.

The 1 ms HHT orbit completes with 1.32803 m position error. HEDOP5 passes at 1 ms (0.30156 mm) and 2 ms (7.90992 mm), but fails at 4 ms (35.3884 mm) against a 10 mm gate. The nonuniform motion near pericenter makes this a demanding fixed-step phase test.

What the result demonstrates

The 4 ms HEDOP5 orbit conserves energy closely but misses the position gate. Refining to 2 ms restores a pass; 1 ms lowers error to 0.302 mm. The result demonstrates sensitivity to pericenter resolution, not a universal safe timestep for orbital dynamics.

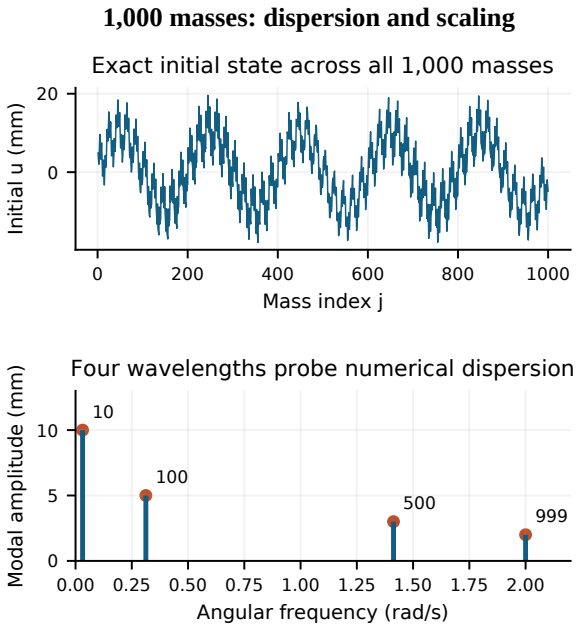


Fig. 10. Exact chain initial displacement and the four excited modal frequencies. Mode labels identify the spatial mode index; all 1,000 masses enter the scored solution.

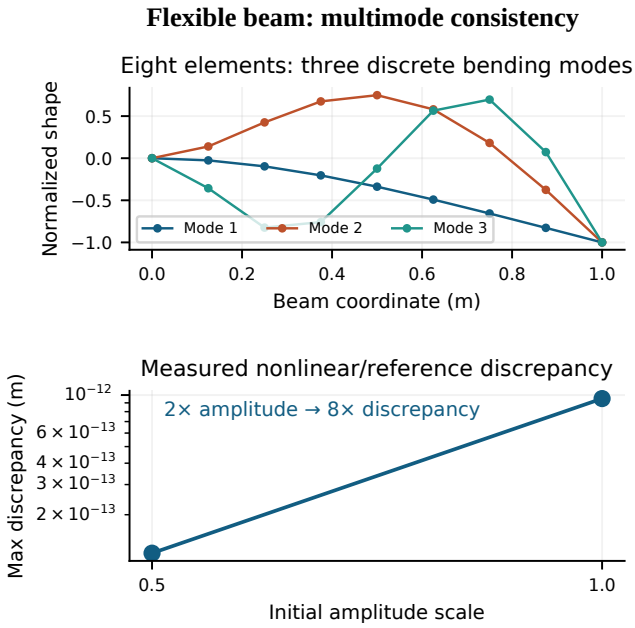


Fig. 11. Linearized discrete beam mode shapes, plus the measured HEDOP5 discrepancy at two amplitude scales. Normalized shapes are illustrative; physical amplitudes are 10, 4 and 2 µm.

Why this case matters

A large DOF count alone is not a useful accuracy test. Exciting four widely separated wavelengths makes the chain sensitive to mode-dependent numerical dispersion and damping. Checking every mass prevents a small set of probes from hiding local error. This measures the cost and fidelity of integrating a large elastic system through the production body/force path, with an exact solution that remains inexpensive to evaluate.

Model and measured evidence

One thousand unit masses are connected by 1,001 springs with stiffness 1 N/m and rest length 1 m, between fixed anchors at x = 0 and 1001 m. Modes p = 10, 100, 500 and 999 have amplitudes 0.01, 0.005, 0.003 and 0.002 m. All initial velocities vanish. The exact longitudinal solution is a modal superposition:

$$u_j(t) = \Sigma_p A_p \sin(\pi jp/1001) \cos(\omega_p t),$$
$$\omega_p = 2 \sin(\pi p/2002).$$

The production model integrates all 1,000 bodies (6,000 velocity DOFs) in the invariant longitudinal subspace. Errors are scored over every mass at saved times, with four independent probe checks.

HHT at 20 ms gives 0.987977 mm error, above the 0.2 mm gate. HEDOP5 at 80 ms gives 0.136698 µm error at 4.297× RT; at 20 ms it gives 1.42563×10^{-10} m at 1.006× RT.

What the result demonstrates

HEDOP5 reaches 4.30× real time at 80 ms with 0.137 µm all-mass error. HHT at 20 ms is fast but fails the displacement gate. This is a controlled longitudinal elastic benchmark; it does not test contact or a long constrained hinge chain.

Why this case matters

A single bending frequency can pass even when a flexible model has incorrect rotary inertia or higher-mode behavior. Three excited modes test the assembled stiffness, lumped masses, rotational inertia and dynamic coupling over a long interval. The comparison isolates temporal integration and agreement with a specified discrete model. It is relevant to flexible robot links and lightweight deployable structures, rather than being a continuum mesh-convergence test.

Model and measured evidence

Eight production Beam6D elements form a 1 m cantilever with nine nodes and one fixed root. The square section is 0.1 m wide, Young's modulus is 1,000 Pa and density is 1,000 kg/m³. Damping is zero. Initial geometry is undeformed; modal velocities excite displacement amplitudes 10, 4 and 2 µm.

The modal reference uses planar Euler–Bernoulli stiffness and the model's lumped translational and rotary masses. Its first three frequencies are 0.1003078301, 0.6027406039 and 1.6060586428 rad/s. Modal evolution is evaluated directly.

The reference is linearized; the production beam is geometrically nonlinear. HHT at 4 ms gives 1.56884×10^{-8} m error. HEDOP5 at 16 ms gives 9.53582×10^{-13} m at 62.463× RT. Halving amplitude reduces discrepancy to 1.19234×10^{-13} m.

What the result demonstrates

The eightfold discrepancy reduction after halving amplitude identifies a finite-amplitude effect rather than an integration error floor alone. The tiny HEDOP5 discrepancy demonstrates agreement with the linearized discrete reference in the small-amplitude regime, not exact nonlinear continuum mechanics.

IV. SELECTED SUCCESSFUL SETTINGS

Case	Step (ms)	Maximum trajectory / reaction error	RT factor
Euler body	4	0.0222071° attitude	69.360×
Lagrange top	4	0.0200826° attitude	55.555×
Near-toggle four-bar	1	0.00357916% peak drive torque	3.097×
Kovalevskaya top	4	8.72905×10^{-8}° attitude	83.095×
Duffing oscillator	4	8.31553×10^{-7} m displacement	38.133×
Near-inverted pendulum	1	0.00244311° bob direction	16.885×
Spherical pendulum	4	4.13072×10^{-5}° bob direction	58.669×
Kepler orbit	1	3.01560×10^{-4} m position	30.281×
1,000-mass chain	80	1.36698×10^{-7} m longitudinal	4.297×
Multimode beam	16	9.53582×10^{-13} m transverse	62.463×

TABLE I. All rows use projected HEDOP5 and complete 1,000 s. Each passes its family-specific accuracy gates. Units and reference types differ, so raw errors are not a common ranking metric. Values are from [2]–[6].

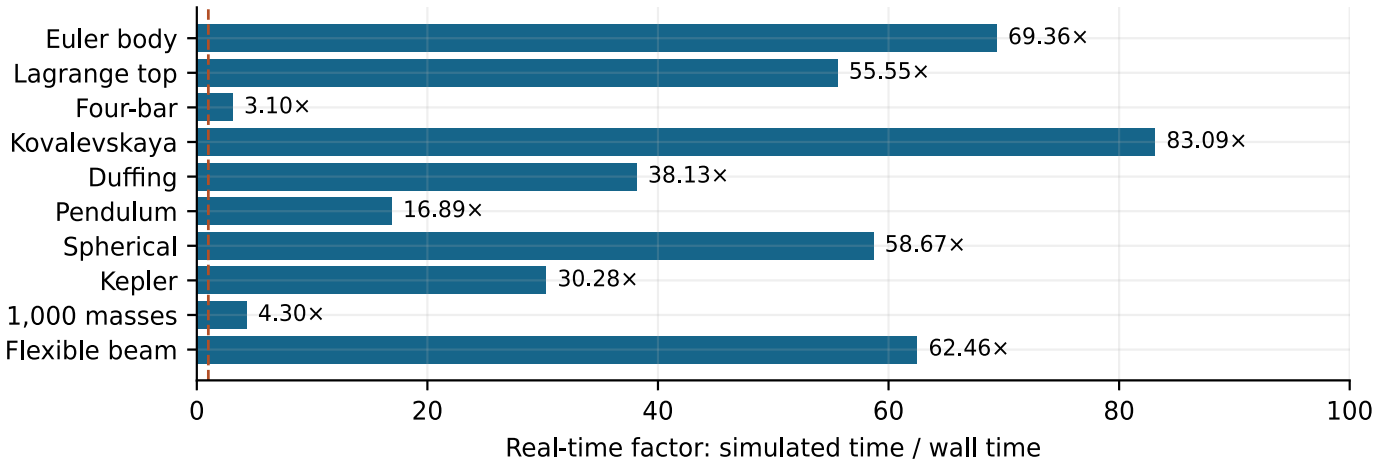


Fig. 12. Measured throughput for the settings in Table I. The dashed line denotes real time (RT = 1). Measurements span different benchmark stages; the figure is not a matched-build speedup comparison.

RT is simulated duration divided by elapsed wall time. It includes initialization, per-step diagnostics, energy calculations and CSV output, and excludes independent reference generation and scoring. These successful settings span 3.097× to 83.095× RT.

The slowest selected case is the near-toggle closed loop. The chain evolves all 1,000 bodies at 4.297× RT. These observations describe the recorded runs; they do not establish a scaling law across models or machines.

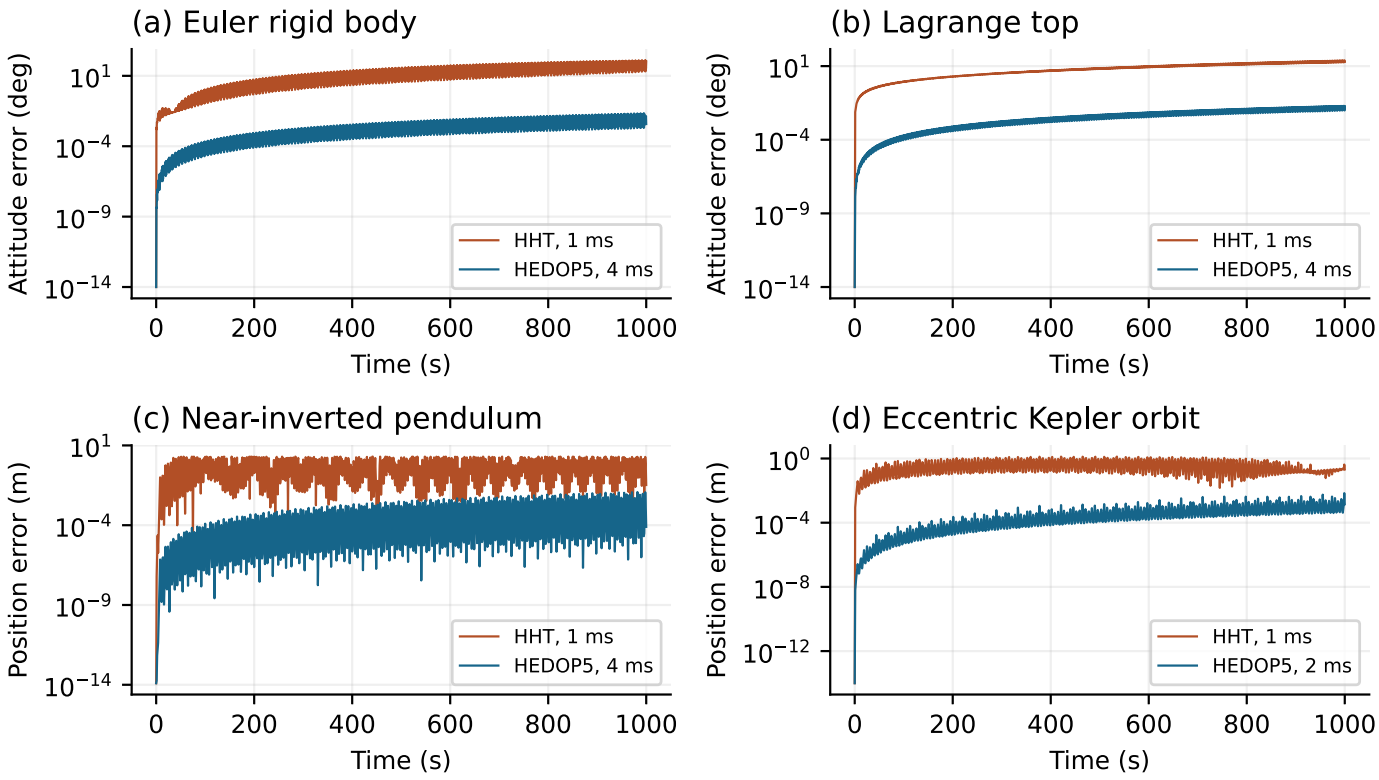


Fig. 13. Recorded compact error histories, without phase alignment or resynchronization. These plots use retained samples; reported maxima were computed before history reduction. The pendulum panel shows position error, while its acceptance gate uses bob-direction error.

V. ACCURACY, FAILURES AND REFINEMENT

Energy error alone is insufficient. The 1 ms HHT Euler run has only 0.002784% maximum energy change but 124.942° attitude error. The corresponding heavy top has 0.04493% energy change and 24.4613° attitude error. HHT passes both after refinement to 0.25 ms. The observations distinguish invariant preservation from accumulated phase accuracy.

The five later families contain 18 completed settings, including refinement and amplitude controls: 13 pass their accuracy gates and five fail. Failures are HHT pendulum, spherical pendulum, Kepler and chain baselines, plus HEDOP5 Kepler at 4 ms. Spherical HHT passes at 0.5 ms; Kepler HEDOP5 passes at 2 and 1 ms. No passing HHT refinement is recorded here for the near-inverted pendulum, Kepler orbit or chain.

Thus “all ten pass” means at least one documented successful setting exists for every family. It does not mean every integrator or timestep passes. The full setting tables remain in the cited evidence.

The near-toggle four-bar isolates a different failure mode. Strict HEDOP5 position projection at 10^{-14} stops at 10.422 s. Relaxing velocity projection alone does not resolve it; relaxing position projection to 10^{-11} does. At the failure time, floating-point spacing of the unwrapped drive angle is 1.42×10^{-14} rad, consistent with a projection roundoff floor.

The 4 ms four-bar rejects a fixed step at 3.452 s. Its cause was not separately diagnosed. Small errors on that partial trajectory cannot establish long-duration accuracy. The successful 1 ms projected run reaches 1,000 s at 3.097× RT. HHT also completes and passes at 1 ms.

For Duffing, all three settings pass. At the same 1 ms step, HHT has greater throughput (35.55× versus 15.59× RT), while HEDOP5 has much smaller displacement error. At 4 ms, HEDOP5 reaches 38.13× RT with submicrometer error. The evidence therefore supports comparing accuracy and cost jointly, rather than selecting a method from step cost alone.

VI. ACCEPTANCE AND REFERENCE QUALIFICATION

The common screen requires completion through the requested endpoint, maximum position-constraint norm below 10^{-8}, velocity-constraint norm below 10^{-4}, and conservative energy change below 1%. Duffing uses 10^{-8} for both constraint residual gates. The externally driven four-bar is excluded from energy-conservation scoring.

Rigid-top attitude and pendulum bob-direction errors must remain below 1°. Four-bar peak-normalized drive-torque error must be below 5%. Duffing displacement must be within 1% of its initial amplitude, with phase error below 1°. Kepler position error must be below 1% of its semimajor axis. Chain and beam displacement errors must be below 1% of the sum of excited modal amplitudes. Kovalevskaya additionally requires relative K drift below 1%. These are screening choices, not local integration tolerances.

Constraint and energy maxima are evaluated at each accepted step. Trajectory errors are maxima over saved times: typically 10–12 ms for the early Euler/top cases, every step for four-bar, 20 ms for Kovalevskaya and 4 ms for Duffing and the later pendulums/orbit. Chain and beam requested sampling intervals are rounded to integer steps. Compact histories are further reduced for plotting; scores precede that reduction.

Independent checks include short ODE comparisons, quadrature checks of elliptic integrals, attitude and COM derivatives, four-bar power balance, eigenrelations, and high-precision special-function evaluation. Duffing residuals are below 3.6×10^{-15}; independent energy agrees with the engine within 6.7×10^{-15} J. Near-inverted pendulum reference checks agree within 1.47×10^{-9} m over 104 samples. A truncated-trajectory negative control is correctly rejected.

VII. REPRODUCIBILITY

Model construction, independent references, scoring scripts, compact error histories and build provenance are available on request. Provenance records identify individual builds; earlier-stage measurements were not remeasured after the cubic/inverse-square spring extensions.

The remaining-five build records Linux x86-64, GCC 14.2.0, C++17, −O2 and −DNDEBUG, with NumPy 2.3.5, SciPy 1.17.0 and pandas 2.2.3. Processor model and thread-affinity details are not supplied in that record. Simulations were timed serially after compilation; light report preparation occurred during runs. Some historical HHT refinements overlapped, so direct speed ratios against those runs are not controlled measurements.

The scripts rebuild every model, run each benchmark and score it against its independent reference; raw trajectories are regenerated rather than stored.

VIII. CONCLUSION

The ten-case study establishes successful 1,000 s settings across all benchmark families, using projected HEDOP5. HHT also provides passing results for the driven four-bar, Duffing oscillator, Kovalevskaya top and beam, and for refined Euler, Lagrange and spherical-pendulum cases. The main finding is that long-duration phase and reaction errors require explicit measurement: completion, small energy drift and tight constraints do not alone guarantee an accurate trajectory. The retained failures and independently qualified references make the suite a reproducible verification asset.

REFERENCES

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